

# Arup Halder

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## EDUCATION

### Dalhousie University

Master of Applied Computer Science, GPA 4.0/4.3

Halifax, NS  
Expected Jan 2027

### Future Institute of Engineering and Management

B.Tech, Computer Science and Engineering, GPA 8.91/10

Kolkata, India  
Jun 2024

## TECHNICAL SKILLS

|              |                                                                                        |
|--------------|----------------------------------------------------------------------------------------|
| Languages    | Java, Python, TypeScript, JavaScript, SQL                                              |
| Backend      | Spring Boot, Spring Security, Spring Data JPA, FastAPI, Node.js, Express.js, REST APIs |
| Frontend     | React, React Native                                                                    |
| Data & Cloud | PostgreSQL, pgvector, MySQL, Flyway, AWS, Docker, Linux, GitHub Actions                |
| ML & Testing | PyTorch, YOLO, OpenCV, ONNX Runtime, JUnit, Testcontainers, Playwright, pytest         |

## EXPERIENCE

### EazyByts

Full Stack Engineer

Kolkata, India  
Jul 2024 – May 2025

- Increased the client's answered-call rate from 40 to 70 percent by building a conversational AI voice agent in **JavaScript** that used natural language processing to qualify callers and push captured data to internal systems through function calls and webhooks.
- Delivered EV charging management across two platforms and four administrative areas by developing web and mobile applications in **React** and **React Native**, backed by Node.js REST APIs.
- Released both EV charging applications with zero critical production defects by securing them with JWT authentication, OAuth sign-in, and role-based access control, and gating each release on unit and integration tests, Postman API validation, and structured code reviews.

### OctaNet Services Pvt. Ltd.

Full Stack Developer

Kolkata, India  
Sep 2023 – May 2024

- Launched a full-stack e-commerce platform supporting a beta rollout of **500+** registered users and **1,200+** product SKUs by engineering a **React** and **TypeScript** storefront integrated with **Node.js**, **Express**, and **MySQL**.
- Achieved zero regressions across core client workflows at deployment, covering **20+** REST API endpoints, by building comprehensive integration test suites in **Postman**.

## PROJECTS

### Dispatch Core · Python, data structures and algorithms, A\*, OpenStreetMap

[GitHub](#)

- Cut nearest-neighbour search cost about 1,144x versus a linear scan (8.74 distance checks per query on 10,000 points) by implementing data structures and algorithms from scratch, a quadtree spatial index and A\* shortest-path search, in **Python** on the Halifax road network graph (3,505 nodes, 8,915 edges).
- Verified algorithm correctness at 98 percent package-wide coverage across 156 tests by validating all output against independent oracles (brute force, networkx, haversine), and reported a routing strategy's 2.4 percent gain as a negative result once it fell within seed-to-seed variance.

### Homebase · Java 21, Spring Boot, PostgreSQL, FastAPI, React, pgvector, Docker

[GitHub](#)

- Enabled housing search, saved listings, and natural-language information retrieval with pgvector in one application by integrating a **React** client with **Java** Spring Boot, FastAPI, and PostgreSQL services.
- Blocked unauthorized access and duplicate saves, verified by 34 CI tests and 25 Playwright end-to-end scenarios, by adding JWT authentication, service-level authorization checks, and a compound database constraint.

### OceanSight Edge · Python, PyTorch, Ultralytics, OpenCV, ONNX Runtime, Streamlit

[GitHub](#)

- Achieved 0.498 mAP50 and 0.274 mAP50-95 on a 108-image holdout by training and benchmarking two YOLO machine learning detectors in **Python** on 1,680 images with source-video-disjoint splits to prevent data leakage.
- Reduced model size by 72 percent (12.14 MB to 3.35 MB) with mAP50-95 moving only from 0.274 to 0.268 by applying selective INT8 quantization for edge deployment on ONNX Runtime.